

In-Class Assignment 1: Simple Linear Regression (SLR) in Practice

Boston Housing: Predicting `medv` with one predictor

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Problem + Data + Variables

- **Goal:** Predict median home value using one predictor at a time (SLR).
- **Dataset:** Boston Housing (R: MASS::Boston).
- **Response:** $Y = \text{medv}$ (median home value, in \$1000s).

Predictors (one per model):

- Model 1: $X_1 = \text{rm}$ (avg rooms per dwelling)
- Model 2: $X_2 = \text{crim}$ (per-capita crime rate)
- Model 3: $X_3 = \text{lstat}$ (% lower status)

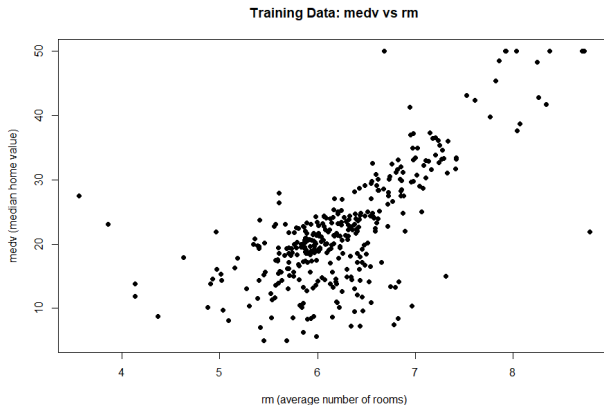
Evaluation Plan (Fair Comparison)

- Split data into **Training (70%)** and **Test (30%)**.
- Record the random seed so results are reproducible.
- **No data leakage:** Outlier rules / transformations decided using training only.
- Compare final models using **Test MSE** (smaller is better).
- Also report **Training R^2** (for interpretation, not ranking).

$$\text{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} (y_i - \hat{y}_i)^2$$

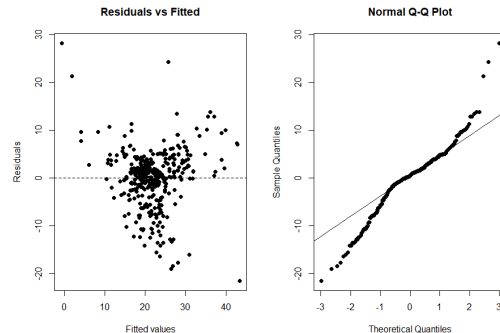
Model 1: $\text{medv} \sim \text{rm}$

Key scatterplot (train):



Final fitted equation: $-30.886 + 8.459 * \text{rm}$

One diagnostic plot:

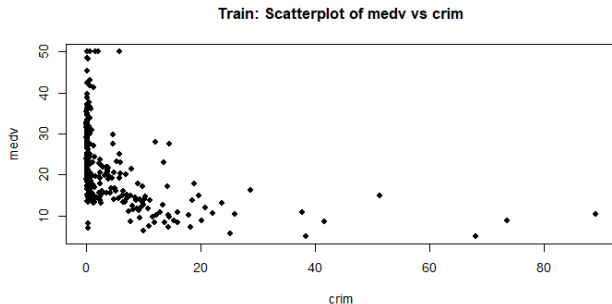


Results (final model):

- Training R^2 : 0.491
- Test MSE: 56.84917

Model 2: medv $\sim$ crim (Outliers Present)

Scatterplot (train):



Fix (training only):

- $1.5 \times \text{IQR}$ rule on crim
- Removed: **52** training points
- Upper bound: **8.3674**

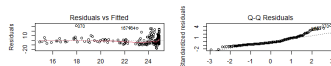
Final equation (cleaned train):

$$\hat{y} = 24.9100 - 1.2264(\text{crim})$$

Results (final model):

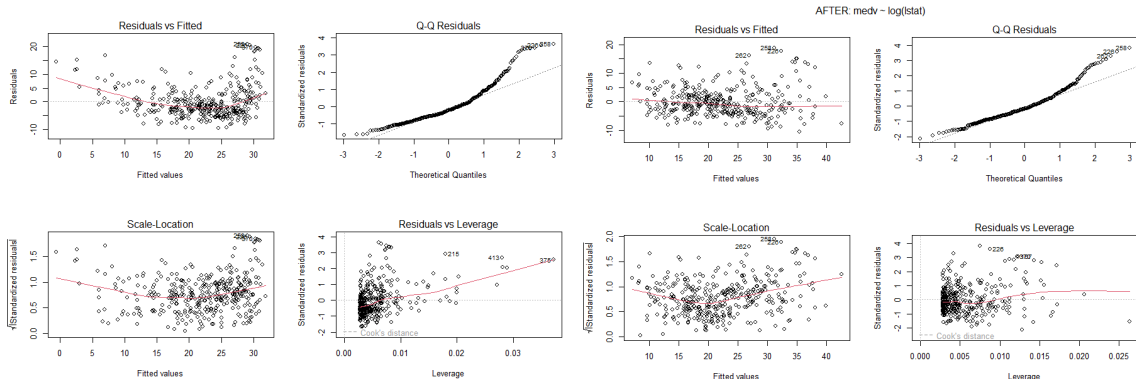
- Training R^2 : **0.0793**
- Test MSE: **112.2664**

One diagnostic plot (cleaned model):



Model 3: $\text{medv} \sim \text{lstat}$ (Transformation Needed)

Untransformed diagnostics show problems: Improved diagnostics (after transform):



Fix (training only): Log transformation

Final equation:

$$50.70 - 11.99 * \log(\text{lstat})$$

Final Comparison + Recommendation

Comparison table (final models):

Model	Predictor	Training R^2	Test MSE	Notes (issues / fix)
3	lstat	0.675	37.74	Nonlinearity/heterosced; used $\log(\text{lstat})$ to improve diagnostics
1	rm	0.491	56.85	Clean SLR; assumptions mostly OK
2	crim	0.0793	112.2664	Outliers in crim; removed 52 train points ($1.5 \times \text{IQR}$)

Ranking (by Test MSE): (fill in best $\rightarrow$ worst)

Recommendation (1–2 sentences):

- Choose the model with the **lowest Test MSE**.
- Mention interpretability (simple story) + performance (test MSE).

Limitations + Next Steps (Optional)

- Only one predictor at a time (SLR can miss important factors).
- Outlier handling / transformations can change the fitted story.
- Next step: multiple regression using more predictors.
- Next step: cross-validation to check stability.

Backup: Definitions (if asked)

- **Outlier:** unusual response relative to the fitted model.
- **Influential point:** a point that noticeably changes the fit if removed.
- **Training R^2 :** fraction of training variability explained by the line.
- **Test MSE:** average squared prediction error on unseen data.